that the universe has nine space dimensions and one time dimension, why is it that three space (and one time) dimensions are large and extended while all of the others are tiny and curled up? Why aren't they all extended, or all curled up, or some other possibility in between? At present no one knows the answer to this question. If string theory is right, we should eventually be able to extract the answer, but as yet our understanding of the theory is not refined enough to reach this goal. That's not to say that there haven't been valiant attempts to explain it. For instance, from a cosmological perspective, we can imagine that all of the dimensions start out being tightly curled up and then, in a big bang-like explosion, three spatial dimensions and one time dimension unfurl and expand to their present large extent while the other spatial dimensions remain small. Rough arguments have been put forward as to why only three space dimensions grow large, as we will discuss in Chapter 14, but it's fair to say that these explanations are only in the formative stages, In what follows, we will assume that all but three space dimensions are curled up, in accordance with what we see around us. A primary goal of modern research is to establish that this assumption emerges from the theory itself.

Third, given the requirement of numerous extra dimensions, is it possible that some are additional time dimensions, as opposed to additional space dimensions? If you think about this for a moment, you will see that it's a truly bizarre possibility. We all have a visceral understanding of what it means for the universe to have multiple space dimensions, since we live in a world in which we constantly deal with a plurality—three. But what would it mean to have multiple times? Would one align with time as we presently experience it psychologically while the other would somehow be "different?"

It gets even stranger when you think about a curled-up time dimension. For instance, if a tiny ant walks around an extra space dimension that is curled up like a circle, it will find itself returning to the same position over and over again as it traverses complete circuits. This holds little mystery as we are familiar with the ability to return, should we so choose, to the same location in space as often as we like. But, if a curled-up dimension is a time dimension, traversing it means returning, after a temporal lapse, to a prior instant in time. This, of